

EDUGENIE — CODE LAYOUT, READABILITY AND REUSABILITY

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FOLDER STRUCTURE

EduGenie/

```
├─ app/
|   ├── __init__.py    (empty)
|   └─ main.py         (all routes + AI logic)
├─ templates/
|   └─ index.html      (single-page UI)
├─ static/
|   └─ style.css        (all styling)
├─ .env
├─ .gitignore
├─ requirements.txt
└─ README.md
```

SEPARATION OF CONCERNS

main.py : All backend logic and routes

index.html : All UI in a single page with 5 cards

style.css : Clean, minimal, distraction-free design

READABILITY PRACTICES

- Shared helper function ``_gen(prompt)`` avoids repetition
- Descriptive route names (explain, summarize, quiz, learn)
- Docstrings on every helper function
- Constants at the top (MODEL_NAME)
- Clean separation of API calls from routing

REUSABILITY

- `_gen()` helper is used by all 5 features
- Each function can be tested independently
- Adding a new feature requires only:
 1. New function using `_gen()`
 2. New route
 3. New card in index.html

NAMING CONVENTIONS

Files : snake_case (main.py)

Functions : snake_case (answer_question)

Routes : lowercase with hyphens or slashes

Constants : UPPER_CASE (MODEL_NAME)

Templates : lowercase (index.html)